

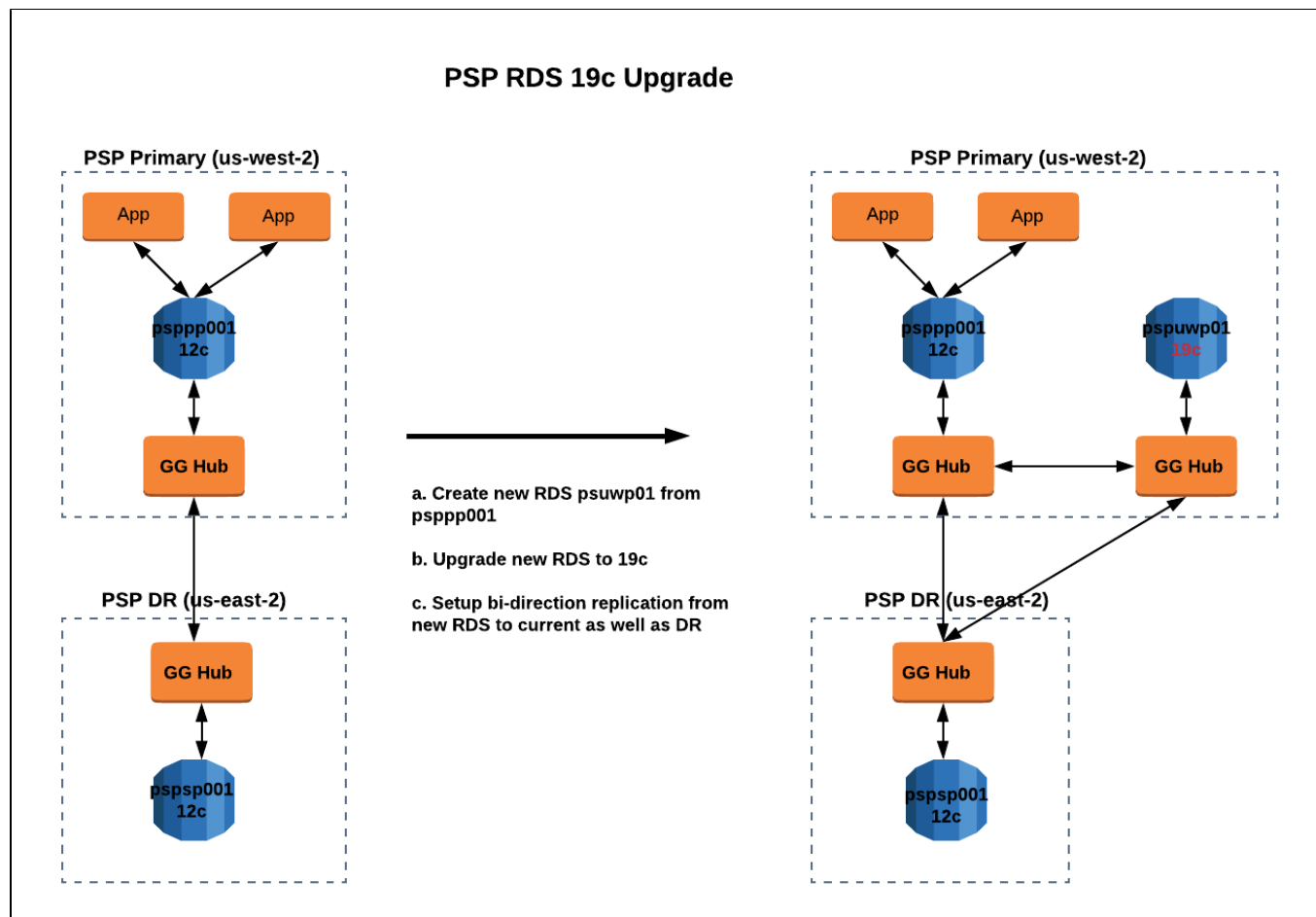
Upgrade PSP RDS with Goldengate

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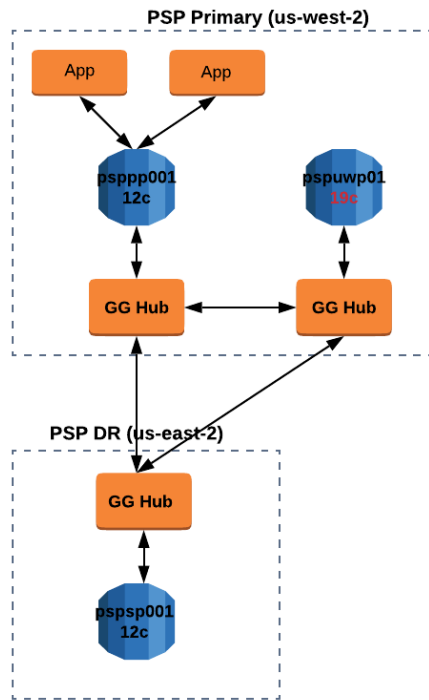
1. Overview

This document will show you a step-by-step on how to upgrade PSP cluster RDS with using Goldengate in the same region. This will require only a 10-15 mins downtime.

The following are PSP RDS 19c Upgrade Process Diagram

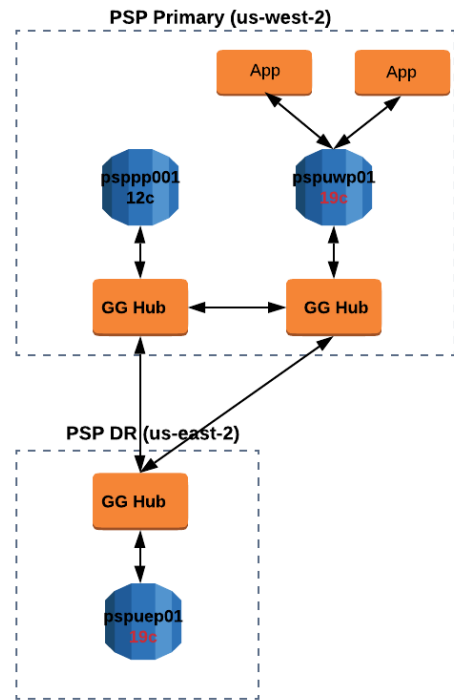


PSP RDS 19c Upgrade

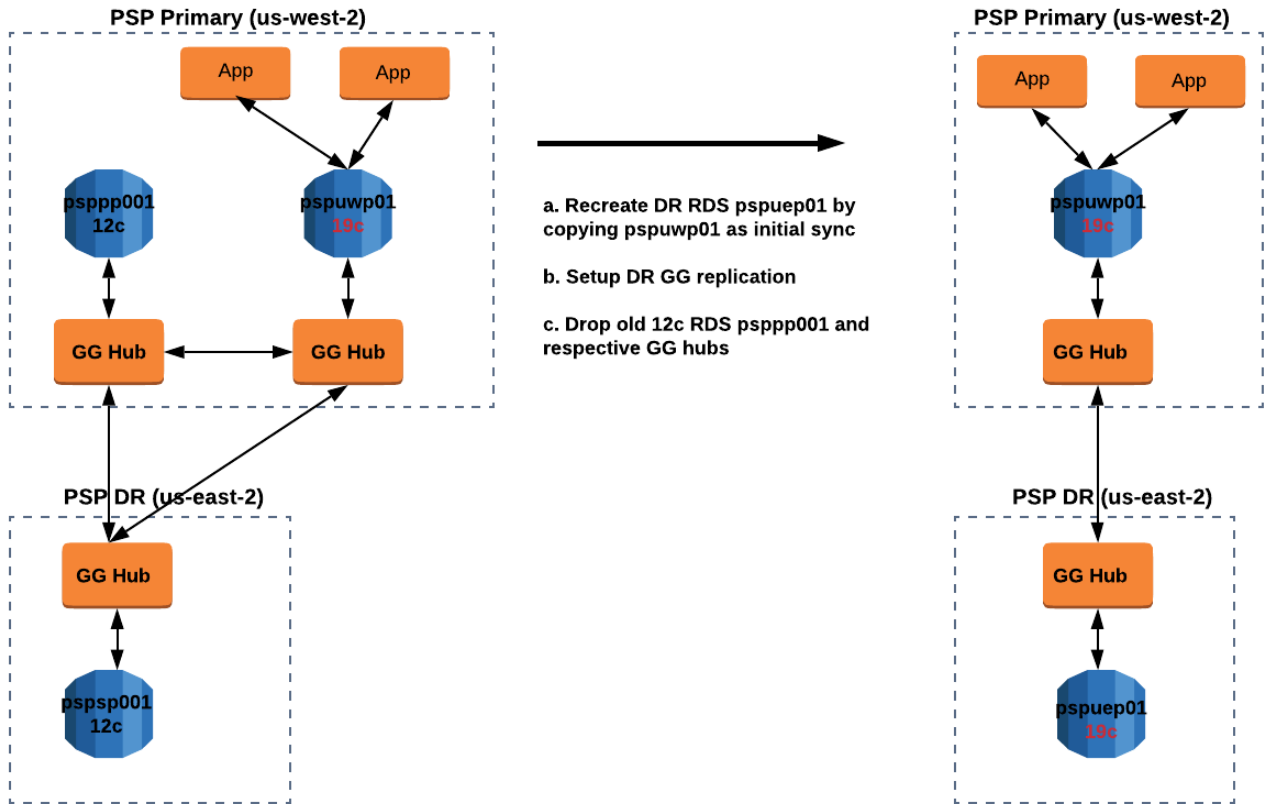


Switchover
(10-15 mins downtime)

- stop application
- ensure GG replication is in sync
- lock app accounts in psppp001
- unlock app accounts in pspuwp01
- start application
- Recreate DR pump and replicat



PSP RDS 19c Upgrade



2. Pre-Setup (Completed)

- Create 19c RDS parameter group and option group

3. Detail Steps

3.1 Naming conversions for production

	(internal) DB Name	DB Identifier	GG Hub Name	Temp OGG Process Example (monolith)	Temp OGG Process Name Trail File Location Example (monolith)
Current	psppp001 pspppib1	psppp001 pspppib1	ggpsppp001 ggpspppib1		
				Extract	
				Pump (to 19c)	ppspwp01 /dirdat/psppp001 /dirdat/psppp001 (remote)

				Replicat (from 19c)	rpswp01 /dirdat/pspuwp01
New	pspuwp01 pspuwp02	pspuwp01 pspuwp02	ggpspuwp01 ggpspuwp02		
				Extract	
				Pump (to 12c)	ppspp001 /dirdat/pspuwp01 /dirdat/pspuwp01 (remote)
				Pump (to AWS us-east)	ppspp001 /dirdat/pspuwp01 /dirdat/pspuwp01 (remote)
				Replicat (from 12c)	rpspp001 /dirdat/ppspp001
				Replicat (from AWS us-east)	rpsps001 /dirdat/pspsp001
12c DR	pspsp001 pspspib1	pspsp001 pspspib1	ggpspsp001 ggpspspib1		
				Pump (to 19c)	ppspwp01 /dirdat/pspsp001 /dirdat/pspsp001 (remote)
				Replicat (from 19c)	rpswp01 /dirdat/pspuwp01
19c DR	pspuep01 pspuep02	pspuep01 pspuep02	ggpspuep01 ggpspuep02		
				Pump (to 19c us-west)	ppspwp01 /dirdat/pspuep01 /dirdat/pspuep01 (remote)
				Replicat (from 19c us-west)	rpswp01 /dirdat/pspuwp01
Old*	psppp001 pspppib1	psppp001 pspppib1	ggpsppp001 ggpspppib1		

3.2 Create new GG Hub in the same region (us-west-2)

- Download the following files https://github.intuit.com/payroll-services-platform/goldengate_hub/tree/master/scripts to your Mac. **Please always re-download the files for each db replication setup.**

```
create_gg_hub_ha.sh
gg_hub_psp.config
goldengate-hub-ha.json
```

- Before you work on production clusters, modify the file goldengate-hub-ha.json as follows because we need m4.4xlarge to support Big Data Adapter

For Monolith:
From: "prod": "m4.xlarge" To: "prod": "m4.10xlarge"

For IBOB:
From: "prod": "m4.xlarge" To: "prod": "m4.4xlarge"

- Create new GG Hub with <DBName> as pspuwp01 (monolith) and pspuwp02 (IBOB) in the same region (us-west-2)

Note:

1) Choose "no" as <HA>

```
chmod +x create_gg_hub_ha.sh
./create_gg_hub_ha.sh
Usage: ./create_gg_hub_ha.sh <AppName> <AppEnv> <DBName> <Region> <VPC> <HA> [<VolumeSize>]
      where <AppName> should be psp, payments, etc
            <AppEnv> should be qa, e2e, stg, sbx, prf or prod
            <Region> should be us-west-2, us-east-2, etc
            <HA> should be yes or no
            <VPC> should be vpc-1, vpc-2, etc or Intuit-vpc-1, Intuit-vpc-2, etc
            [<VolumeSize>] is optional. It is EBS volume /u01 size in GB and default is 1000 GB
```

For prod, the command should like below,
./create_gg_hub_ha.sh psp prd pspuwp01 us-west-2 vpc-2 no

For preprod, the command should like below,
./create_gg_hub_ha.sh psp prf psppf501 us-west-2 vpc-2 no

- Download upgrade scripts/files on both GG Hubs by running the following commands

```
. /l/orcl
mkdir -p /u01/upgrade/
cd /u01/upgrade/
if [ `hostname |cut -d'.' -f1 |grep 'ppp' |wc -l` -gt 0 ]; then
    aws s3 cp s3://bits-sbg-ppsp-${VPC_ENV}-us-west-2/binaries/goldengate/upgrade/scripts_on_current.tar .
else
    aws s3 cp s3://bits-sbg-ppsp-${VPC_ENV}-us-west-2/binaries/goldengate/upgrade/scripts_on_new.tar .
fi
tar -xvf *tar
chmod +x */*.sh
```

- Copy temp gg replication files for pump and replicat on new GG hub

```
. /l/orcl
cd /u01/upgrade/scripts_on_new

For monolith:
cp prm_obey_files/monolith/*.prm $GG_HOME/dirprm/
cp prm_obey_files/monolith/*.oby $GG_HOME/diroby/

mkdir -p $GG_HOME/dirdat/${new_rds_sid}

For IBOB:
cp prm_obey_files/ibob/*.prm $GG_HOME/dirprm/
cp prm_obey_files/ibob/*.oby $GG_HOME/diroby/

mkdir -p $GG_HOME/dirdat/${new_rds_sid}
```

- check and review all newly created parameter, obey files.
- Setup mgr on the new GG Hub

```
cd /u01/upgrade/scripts_on_new
./01_configure_goldengate_n.sh |tee 01_configure_goldengate_n.log
```

- Create temporary pump on the current GG Hub

```
. /l/orcl
cd /u01/upgrade/scripts_on_current

For monolith:
cp prm_obey_files/monolith/*.prm $GG_HOME/dirprm/
cp prm_obey_files/monolith/*.oby $GG_HOME/diroby/

mkdir -p $GG_HOME/dirdat/${new_rds_sid}

For IB0B:
cp prm_obey_files/ibob/*.prm $GG_HOME/dirprm/
cp prm_obey_files/ibob/*.oby $GG_HOME/diroby/

mkdir -p $GG_HOME/dirdat/${new_rds_sid}
```

- start pump on current GG hub. this pump would be running temporarily until switchover

```
get last trail sequence number from DR pump info
info <dr_pump>

./ggsci
obey diroby/create_${pump_to_new_hub}.oby
alter extract ${pump_to_new_hub} extseqno $last_trail_seqno, extrba 0
start ${pump_to_new_hub}
exit
```

- Verify temporary pump on both GG Hubs

```
On the current GG Hub:
ggsci
info all

On the new GG Hub:
ls -trl $GG_HOME/dirdat/*pp001*/
```

3.3 Create new RDS from current RDS

- Run the following command from local mac laptop

```
mchoubey@INTUL1748e09cd$ ./copy_rds_cross_region.sh
Usage: ./copy_rds_cross_region.sh <AppEnv> <DBType> <Source DBName> <Target DBName> <Source Region> <Target
Region> <Target VPC> [<DBInstanceClass>]
    where <AppEnv> should be qa, e2e, stg, sbx, prf, prod
           prodn means new prod (with less half of regular prod storage)
           <DBType> should be clusterdb or reportdb
           <Source DBName> and <Target DBName> should be us-west-2 or us-east-2
           <Target VPC> should be vpc-1, vpc-2, etc
           <DBInstanceClass> should be db instance class like db.r4.xlarge. It is optional with the default db.
r4.xlarge

For prod, the command should like below
nohup ./copy_rds_cross_region.sh qa clusterdb pspuatuw pspsys01 us-west-2 us-west-2 vpc-2 db.r5.2xlarge>
copy_rds_cross_region_pspuatuw.log &

For preprod, the command should like below
nohup ./copy_rds_cross_region.sh qa clusterdb pspuatuw pspsys01 us-west-2 us-west-2 vpc-2 db.r5.2xlarge>
copy_rds_cross_region_pspuatuw.log &
```

3.4 Upgrade the new RDS

- Run the following command from local mac laptop

```
mchoubey@INTUL1748e09cd$ ./upgrade_rds.sh
./upgrade_rds.sh <profile> <region> <DBName> <NewVersion>
where <Profile> is AWS account profile. 'no' means no profile provided
    <Region> is AWS region
    <DBName> is RDS name
    <NewVersion> is the RDS version (like 19.0.0.0.ru-2020-01.rur-2020-01.r1) you want to upgrade to

The command should like below
nohup ./upgrade_rds.sh sbg-psp-ppd us-west-2 pspsysib 19.0.0.0.ru-2020-01.rur-2020-01.r1 > upgrade_rds_pspsysib.
log &
```

- **Lock all app user accounts**

```
alter user PSPAPP account lock;
alter user PSPADM account lock;
```

- Run the following command on the new GG Hub to confirm logging to 19c DB. Make sure seeing "Oracle Database 19c"

```
[oracle@ggpspibob1 scripts_on_new]$ ./login_n.sh

SQL*Plus: Release 12.1.0.2.0 Production on Thu Jul 2 04:50:15 2020

Copyright (c) 1982, 2014, Oracle. All rights reserved.

Enter password:
Last Successful login time: Thu Jul 02 2020 04:49:50 -07:00

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production

SQL> Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
```

3.5 Create replicat on the new GG Hub

- Run the following command on the new GG Hub to confirm logging to 19c DB. Make sure seeing "Oracle Database 19c"

```
[oracle@ggppspibob1 scripts_on_new]$ ./login_n.sh

SQL*Plus: Release 12.1.0.2.0 Production on Thu Jul 2 04:50:15 2020

Copyright (c) 1982, 2014, Oracle. All rights reserved.

Enter password:
Last Successful login time: Thu Jul 02 2020 04:49:50 -07:00

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production

SQL> Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
```

- Create and start replicat on the new GG Hub

```
cd /u01/upgrade/scripts_on_new
./02_configure_goldengate_n.sh |tee 02_configure_goldengate_n.log

This script will have the following prompts:

Please enter the database master user intuadmin password:
Please enter the database GGS/GGT password:

(after all db-related Goldengate configuration steps is completed and replicat process is created. Ensure
script has used right obey and parameter file to create replicat.)
Do you want to start the replicat at scn <a real scn> now (y/n)?
<Enter "y" if the output from the script looks good; otherwise, enter "n".
For example, if wrong password was entered, you should enter "n" and then rerun the script>
```

- Run the following commands to verify the replicat is running on the new GG Hub

```
. /l/orcl
ogg
ggsci

info all
info <new replicat>

send <new replicat> stats
like
send RPSPP001 stats
You should see status on some tables
```

- verify GG replication sync. Create/update gg_heartbeat table under pspadm schema.

```
SQL> alter session set nls_date_format='yyyy-mm-dd hh24:mi:ss';
Session altered.
SQL> select * from pspadm.gg_heartbeat;
SOURCE                LAST_UPDATE
-----
<RDS_SID>_12c          2017-02-11 11:52:00

Make sure that the record <RDS_SID>_12c is close to current time (the difference should not be great than 1
minute)
```

- Run the following scripts to verify sequence replication

On Source and target :

```
set pagesize 1000
set numwidth 30
col SEQUENCE_OWNER format a10
col SEQUENCE_NAME format a32
select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQUENCE_OWNER in ('PSPADM') order
by SEQUENCE_OWNER, SEQUENCE_NAME;
```

The output will look like below and #s on target will be always not smaller than #s on source

3.7 Setup Veridata comparison job

- Follow [Setup veridata agent](#) to setup Veridata agent for new RDS.
- schedule veridata compare between 12c and 19c RDS on veridata server.

3.8 Setup replication from the new RDS to the current RDS

- Run the following command on the new GG Hub to confirm logging to 19c DB. Make sure seeing "Oracle Database 19c"

```
[oracle@ggppibob1 scripts_on_new]$ ./login_n.sh

SQL*Plus: Release 12.1.0.2.0 Production on Thu Jul 2 04:50:15 2020

Copyright (c) 1982, 2014, Oracle. All rights reserved.

Enter password:
Last Successful login time: Thu Jul 02 2020 04:49:50 -07:00

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production

SQL> Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
```

- Create and start extract and pump on the new GG Hub.

```
cd /u01/upgrade/scripts_on_new
./configure_goldengate_on_primary_n.sh |tee configure_goldengate_on_primary_n.log
```

- review script execution log for any error.
- Verify temporary pump on both GG Hubs

On the new GG Hub:

```
ggsci
info all
```

On the current GG Hub:

```
ls -trl $GG_HOME/dirdat/*uw*/
```

- Create replicat on the current GG Hub

verify trail file sequence number from above step and use that

```
./ggsci
obey diroby/create_${tmp_replicat}.oby
obey ./diroby/dblogin_t.oby
add checkpointtable
alter replicat ${tmp_replicat} extseqno $trail_seqno, extrba 0
start ${tmp_replicat}
info all
exit
```

- Verify temporary replicat on the current GG Hub

On the new GG Hub:
ggsci

```
info all
info <temp replicat>

send <temp replicat> stats
like
send rpswp01 stats
You should see status on PSPADM.GG_HEARTBEAT
```

- Setup monitor for new GG process on the new GG Hub
- Setup monitor for new GG process on the current GG Hub
- Enable cron job on the new GG Hub

```
crontab /u01/ogg/scripts/crontab.txt
```

3.9 Setup replication from new gg hub to DR

- Create DR pump on the new GG Hub. – **double check/review**

get last trail sequence number from EXTTRAILSOURCE (for eg. ./dirdat/pspuwp01/tr for monolith) of extract.

```
ls -ltr ./dirdat/pspuwp01/tr*|tail
```

```
./ggsci
obey diroby/create_${pump_to_dr}.oby
alter extract ${pump_to_dr} extseqno $last_trail_seqno, extrba 0
info all
start ${pump_to_dr}
exit
EOF
```

- Create and start replicat on the **DR GG Hub**

verify trail file sequence number from above step and use that

```
./ggsci
obey diroby/create_${dr_replicat}.oby
obey ./diroby/dblogin_t.oby
add checkpointtable
start ${dr_replicat}
info all
exit
```

```
ggsci
info <dr_replicat>
send <dr_replicat> stats
```

Note: Alter replicat to correct sequence number as below (if required)
alter replicat \${dr_replicat} extseqno \$trail_seqno, extrba 0

- Verify replicat on the DR GG Hub

```
ggsci
info <replicat>
send <replicat> stats
```

- set up veridata compare job between 19c and 12c DR RDS databases after switchover.

3.10 Switchover application

- (Pre-Switchover) Create a CR for Switchover at least one day in advance
- (Pre-Switchover) Make sure that the Veridata report in last 24 hours is clean
- Check GG processes on both **both current and new GG Hubs** to make sure there is no lag

1. info all

2. Check lag via heartbeat on Monolith and IB0B
login to 19c & 12c

Monolith :
ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from pspadm.gg_heartbeat;

IB0B :
ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from ibobadm.gg_heartbeat;

- **(Downtime Begin)** shut down application
- Verify connection on the current GG Hub

```
cd /u01/upgrade/scripts_on_current
./login_p.sh
```

```
@chk_session_countg.sql
```

- Lock application user accounts on current database (12c)

```
1.
Monolith :
alter user PSPADM account lock;
alter user PSPAPP account lock;

/* This is for FYI
alter user INTU_IQUATE account lock;
alter user OIMUSER account lock;
alter user PSPREAD account lock;
alter user PSPLOG account lock;
alter user PSPREPORTS account lock;
alter user APPBUGFIX account lock;
alter user PSPBOADMIN account lock;
alter user PSPFLUXADM account lock;
alter user PSPFLUXADMIN08 account lock;
alter user PSPREAD account lock;
alter user PSPTEMP account lock;
*/

IBOB :
alter user IBOBADM account lock;
alter user IBOB_PROD_PSPAPP account lock;
alter user IBOB_PROD_PSPREAD account lock;

2. Kill existing connections (if still exist)
```

- Verify account_status is LOCKED

```
Monolith:
select username,account_status from dba_users where username in ('PSPADM','PSPAPP');

IBOB:
select username,account_status from dba_users where username like '%IBOB%';
```

- Verify GG replication

```
1. info all

2. Check lag via heartbeat on Monolith and IBOB
login to 19c & 12c

Monolith :
ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from pspadm.gg_heartbeat;

IBOB :
ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from ibobadm.gg_heartbeat;
```

- restart extract EPSP001 on current GG hub

- Unlock application user accounts on 19c database

```
1.
Monolith :
alter user PSPADM account unlock;
alter user PSPAPP account unlock;

/* This is for FYI
alter user INTU_IQUATE account unlock;
alter user OIMUSER account unlock;
alter user PSPREAD account unlock;
alter user PSPLOG account unlock;
alter user PSPREPORTS account unlock;
alter user APPBUGFIX account unlock;
alter user PSPBOADMIN account unlock;
*/

IBOB:
alter user ibobadm account unlock;
alter user IBOB_PROD_PSPAPP account unlock;
alter user IBOB_PROD_PSPREAD account unlock;
```

- verify accounts are unlocked

```
Monolith:
select username,account_status from dba_users where username in ('PSPADM','PSPAPP');

IBOB:
select username,account_status from dba_users where username like '%IBOB%';
```

- **(Downtime End)** Start application
- Verify connection on the new GG Hub

```
cd /u01/upgrade/scripts_on_new
./login_n.sh

@chk_session_countg.sql
```

3.11 Cleanup 12c infra and GG processes

When it is ready to drop the old 12c RDS (normally one week after switchover), proceed the following step to drop the old 12c GG Hub, RDS and cleanup GG processes on new GG Hub

- Drop 12c pump and replicat on new gg hub. These pump and replicat created for replicating changes from 12c RDS.

```
cd $GG_HOME
./ggsci
info all
obey ./diroby/dblogin_t.oby
delete $tmp_pump
delete $tmp_replicat
info all
exit
```

Note: Please verify and then delete the temp trail file directory \$GG_HOME/dirdat/\${tmp_rds_sid}

- Delete the temp trail file directory as prompted from the step above
- Make sure auto_start.flag exists and crontab, otherwise execute as below

```
touch /u01/ogg/scripts/auto_start.flag
crontab /u01/ogg/scripts/crontab.txt
```

3.12 Setup cross-region Goldengate replication

- Follow [Setup PSP DB Cross-region Replication with Goldengate after RDS 19c Upgrade](#)

4. Appendix

4.1 (ROLLBACK) Switchback Application

- Check GG processes on both GG Hubs to make sure there is no lag

```
info all
```

- **(Downtime Begin)** Put offline page, Stop all JVMs and any other jobs
- Verify connection on the new GG Hub

```
cd /u01/upgrade/scripts_on_new
./login_p.sh
```

```
@chk_session_countg.sql
```

- Lock application user accounts on current database **(19c)**

```
1.
Monolith :
alter user PSPADM account lock;
alter user PSPAPP account lock;

/* This is for FYI
alter user INTU_IQUATE account lock;
alter user OIMUSER account lock;
alter user PSPREAD account lock;
alter user PSPLOG account lock;
alter user PSPREPORTS account lock;
alter user APPBUGFIX account lock;
alter user PSPBOADMIN account lock;
alter user PSPFLUXADM account lock;
alter user PSPFLUXADMIN08 account lock;
alter user PSPREAD account lock;
alter user PSPTEMP account lock;
*/

IBOB :
alter user IBOBADM account lock;
alter user IBOB_PROD_PSPAPP account lock;
alter user IBOB_PROD_PSPREAD account lock;
```

```
2. Kill existing connections (if still exist)
```

- Verify account_status is LOCKED

```
Monolith:
select username,account_status from dba_users where username in ('PSPADM','PSPAPP');

IBOB:
select username,account_status from dba_users where username like '%IBOB%';
```

- Verify GG replication

```
1. info all

2. Check lag via heartbeat on Monolith and IBOB
login to 12c

Monolith :
ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from pspadm.gg_heartbeat;

IBOB :
ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from ibobadm.gg_heartbeat;
```

- restart extract EPSPWP01 on new GG hub
- Unlock application user accounts on 12c database

```
1.
Monolith :
alter user PSPADM account unlock;
alter user PSPAPP account unlock;

/* This is for FYI
alter user INTU_IQUATE account unlock;
alter user OIMUSER account unlock;
alter user PSPREAD account unlock;
alter user PSPLOG account unlock;
alter user PSPREPORTS account unlock;
alter user APPBUGFIX account unlock;
alter user PSPBOADMIN account unlock;
*/

IBOB:
alter user ibobadm account unlock;
alter user IBOB_PROD_PSPAPP account unlock;
alter user IBOB_PROD_PSPREAD account unlock;
```

- verify accounts are unlocked

```
Monolith:
select username,account_status from dba_users where username in ('PSPADM','PSPAPP');

IBOB:
select username,account_status from dba_users where username like '%IBOB%';
```

- **(Downtime End)** start application
- **Verify connection on the current GG Hub**

```
cd /u01/upgrade/scripts_on_current  
./login_n.sh
```

```
@chk_session_countg.sql
```